Report

Of

# AI/ML VIRTUAL INTERNSHIP:

# CLASSIFICATION AND PREDICTION SYSTEMS USING MACHINE LEARNING

Submitted

To

**School of Computer Science and Engineering**

**IILM University, Greater Noida, U.P.**

In partial fulfilment of the requirement of degree of

**B.Tech CSE**

By


Roll Number of Student: 25SCS1003002304

**Name of Student: BAPPADITYA DAS**

Batch: 2026-27

# CANDIDATE'S DECLARATION

I, **BAPPADITYA DAS**, Roll No. 25SCS1003002304, do hereby declare that the internship report titled **“AI/ML Virtual Internship: Classification and Prediction Systems Using Machine Learning”** has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in CSE. All the references have been quoted and I have not taken as such any material from any other source. I affirm to you that this report is my own and has not been submitted to any other institute for any degree/diploma requirement and further I shall be solely responsible for any kind of copyright violation in this regard.

Bappaditya

Signature
Student Name: BAPPADITYA DAS
Roll No.: 25SCS1003002304

# ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their aspiring guidance, invaluably constructive criticism and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work.

I would like to thank SmartED Innovations for providing the opportunity to undergo this Virtual Internship in Artificial Intelligence and Machine Learning, and for the training and resources that helped me build the projects discussed in this report. I would also like to thank the faculty of the School of Computer Science and Engineering, IILM University, Greater Noida, for their continuous support and encouragement throughout this internship.

I express my gratitude to my parents for their blessings.

Bappaditya

Signature

Student Name: BAPPADITYA DAS
Roll No.: 25SCS1003002304

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## Internship Completion Certificate

The internship offer letter and the certificate of completion issued by SmartED Innovations on successful completion of the Virtual Internship in Artificial Intelligence are attached below.

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**Date:** 02-June-2026

To,
Bappaditya Das
IILM University, Greater Noida

**Subject:** Internship Offer – **Artifical Intelligence & Machine Learning**

Dear **Bappaditya Das**,

We are pleased to offer you a Virtual Internship in **Artifical Intelligence & Machine Learning** with **SmartED Innovations**. This program is designed to provide practical training and valuable professional exposure.

**Internship Details:**
**Institution:** IILM University, Greater Noida
**Mode:** Virtual Mode
**Start Date:** 02-June-2026

**Terms & Conditions:**
**Status & Compensation:** This is a training internship. You will not hold employee status and will not receive salary, stipends, or benefits. Completion of this internship does not guarantee employment.
**Confidentiality:** You must maintain strict confidentiality of all Company information and return/delete any company property or data at the end of the internship.
**Policies:** By accepting, you agree to follow all applicable Company policies and Terms and Conditions.
**Agreement:** This letter represents the full understanding between you and the Company. Any modification must be in writing and signed by both parties.

We look forward to working with you and believe this internship will support your academic and professional growth. Please confirm acceptance by Replying **"I ACCEPT THE OFFER"**.

If you have any questions, please do not hesitate to contact us. Very truly yours,

Nagiri Gokul
Managing Director
**SmartED Innovations**

SMARTED

Note: This is a computer-generated letter. No signature is required

**SmartED Innovations**
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4thFloor,KS ARCADE building, 167, 9th Main Rd, 6th sector Bengaluru, Karnataka,560102.
info@smarted.pro , operations@smarted.pro +91-7892227891 ,

**Annexure I: Internship Offer Letter**


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CERTIFICATE
OF INTERNSHIP COMPLETION

Bappaditya Das

Has successfully completed training and an internship with **SmartED Innovations** in,
Artificial Intelligence

Throughout this internship, the student has consistently shown exceptional enthusiasm, dedication, and a strong willingness to learn and contribute

**STUDENT ID:** AIIC1327

**Date:** 11 Jul 2026

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SAI SREEKAR
Director & co-founder

USNG LEARNSPACE EDTECH PVT LTD www.smarted.pro

**Annexure II: Certificate of Internship Completion**

# 4. Project Description

## 4.1 Introduction

This report documents the work undertaken during a Virtual Internship in Artificial Intelligence and Machine Learning, completed with SmartED Innovations in association with IILM University, Greater Noida, as part of the academic requirement for the 3rd semester B.Tech CSE curriculum. The internship combined theoretical instruction, delivered across six structured modules, with hands-on, project-based learning. As part of the practical component, two independent Machine Learning based systems were designed and implemented: (i) a classification system to categorise New York City (NYC) room/listing data, and (ii) a predictive system to estimate a Mental Health Score from a set of behavioural and lifestyle indicators. Both projects were implemented in Python using standard data-science libraries, and are described in detail in the sections that follow. A module-wise summary of the theoretical curriculum covered during the internship is presented in Section 4.10 (Internship Training Log Book).

## 4.2 Organization Profile

SmartED Innovations is a Bengaluru-based ed-tech organisation that partners with universities to deliver industry-oriented virtual internship programs to students across emerging technology domains such as Artificial Intelligence, Data Science, and Web Development. The organisation's programs are structured to combine conceptual/theory sessions with guided, hands-on project work, enabling students to apply classroom learning to practical problems while working remotely. The internship undertaken for this report was carried out entirely in Virtual Mode, under this arrangement between SmartED Innovations and IILM University, and ran from 26 October 2025 to 11 July 2026.

## 4.3 Problem Statement

The internship's practical component was centred on two distinct data-driven problems:

- NYC Room Classification: Given a dataset of room/property listings from New York City with attributes such as location (borough/neighbourhood), room type, price, minimum nights, and availability, the task was to build a classification model capable of categorising listings (for example, by room type or price category) based on these attributes.

- Mental Health Score Prediction: Given a dataset of lifestyle and behavioural indicators (such as sleep patterns, screen time, physical activity, and stress-related factors), the task was to build a predictive model capable of estimating a Mental Health Score for an individual, so that key contributing factors could be identified.

Both problems are representative of common real-world applications of supervised Machine Learning: the first is a classification task, and the second is a regression/score-prediction task.

### 4.4 Project Objectives

- To understand and apply the end-to-end Machine Learning workflow, from raw data to a working predictive model.
- To perform data cleaning, exploratory data analysis (EDA), and feature engineering on real-world style datasets.
- To implement and compare classification algorithms for the NYC room classification task.
- To implement a regression-based model for the mental health score prediction task and interpret which factors influence the predicted score.
- To evaluate model performance using appropriate metrics and visualise results for interpretability.
- To gain practical, hands-on exposure to Python's data-science ecosystem (Pandas, NumPy, Scikit-learn, Matplotlib/Seaborn).

### 4.5 Scope of the Project

The scope of both projects is limited to an educational, proof-of-concept implementation using publicly available/sample datasets, and is not intended for production deployment. The NYC room classification model is restricted to the features present in the dataset used and does not incorporate real-time listing data. Similarly, the mental health score prediction system is intended to demonstrate how lifestyle indicators can be modelled statistically to estimate a wellness-related score, and is not a diagnostic or clinical tool. The focus throughout was on understanding and correctly applying the Machine Learning pipeline rather than on achieving production-grade accuracy.

## 4.6 Technologies and Tools Used

- Programming Language: Python 3
- Data Handling: Pandas, NumPy
- Machine Learning: Scikit-learn (classification and regression models, train-test split, evaluation metrics)
- Data Visualisation: Matplotlib, Seaborn
- Development Environment: Jupyter Notebook / Google Colab
- Version of tools were kept consistent with the standard Anaconda/Python data-science distribution used during the internship.

## 4.7 System Architecture

Both systems follow the standard supervised Machine Learning pipeline shown in Figure 4.1: raw data is collected, cleaned and preprocessed, relevant features are engineered/selected, a model is trained on the processed data, the trained model is evaluated on unseen (test) data, and finally used to generate predictions on new inputs.

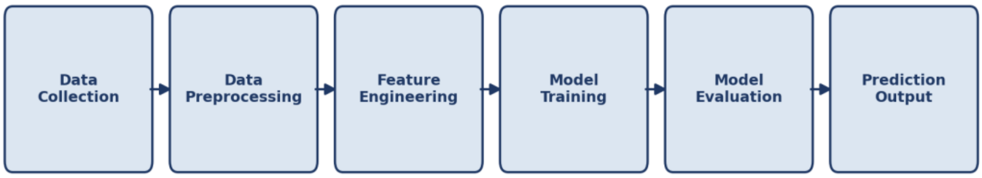



**Fig. 4.1: General Machine Learning Pipeline followed for both projects**

## 4.8 Methodology

### 4.8.1 NYC Room Classification

- Data Loading & Cleaning: The dataset was loaded using Pandas; missing values (e.g., in the reviews-per-month or name fields) were handled by imputation or row removal as appropriate, and irrelevant/high-cardinality identifier columns were dropped.

- Exploratory Data Analysis: Distribution of listings by borough, room type, and price range was visualised to understand the dataset before modelling.
- Feature Encoding: Categorical features such as neighbourhood group and room type were encoded (Label Encoding/One-Hot Encoding) for use in the model.
- Train-Test Split: The dataset was split into training and testing subsets (typically 80:20) using Scikit-learn's train_test_split.
- Model Training: Classification algorithms such as Logistic Regression, Decision Tree, and Random Forest Classifier were trained on the processed training data.
- Evaluation: Model performance was assessed using accuracy score, confusion matrix, and precision/recall for the classification task.

### 4.8.2 Mental Health Score Prediction

- Data Loading & Cleaning: The dataset containing lifestyle/behavioural indicators (e.g., sleep hours, screen time, exercise frequency, social interaction) was loaded and cleaned for missing/inconsistent entries.
- Feature Scaling: Numerical features were scaled/normalised where required to ensure comparability across different value ranges.
- Train-Test Split: The data was split into training and testing sets for unbiased evaluation.
- Model Training: Regression-based models (such as Linear Regression and Random Forest Regressor) were trained to predict a continuous Mental Health Score from the input features.
- Evaluation: Model performance was evaluated using metrics such as Mean Absolute Error (MAE), Mean Squared Error (MSE), and the $R^2$ score, and feature importance was analysed to identify the most influential lifestyle factors.

## 4.9 Expected Outcomes

- A working classification model that can categorise NYC room listings with reasonable accuracy based on the chosen features.

- A working regression model that can estimate a Mental Health Score from lifestyle indicators, along with insight into which factors most strongly influence the predicted score.
- Practical, transferable familiarity with the full Machine Learning workflow — data cleaning, EDA, feature engineering, model training, and evaluation — using industry-standard Python tools.
- A foundation for further exploration of more advanced algorithms and larger, real-world datasets in future coursework or projects.

### 4.10 Internship Training Log Book

The internship curriculum was delivered through six structured, self-paced theory modules covering the foundations of Artificial Intelligence through to modern Generative AI applications, in addition to the two hands-on projects described above. Table 4.1 summarises the key topics covered under each module.

**Table 4.1: Internship Training Log Book – Module-wise Summary**

| Module | Module Title | Key Topics Covered | Key Learning Outcome |
|---|---|---|---|
| Module 1 | Introduction to AI and Python Essentials | • What is AI? From Turing Test to Today<br>• Types of AI: Capabilities, Direction, and Technology<br>• Python Essentials: A Beginner's Guide<br>• Working with NumPy and Pandas<br>• Mathematical Foundations: Vectors, Matrices & Probability Concepts<br>• Statistics: Mean, Median, Mode, Variance, Standard Deviation | Built foundational understanding of AI and developed the Python and mathematical/statistical skills needed for data science work. |
| Module 2 | Problem Solving Agents and Search Algorithms | • Agents & Rationality: Understanding Intelligent System Design<br>• PEAS Framework & Environment Types in AI | Learned to formulate problems as search tasks and apply uninformed and informed search strategies to solve them. |

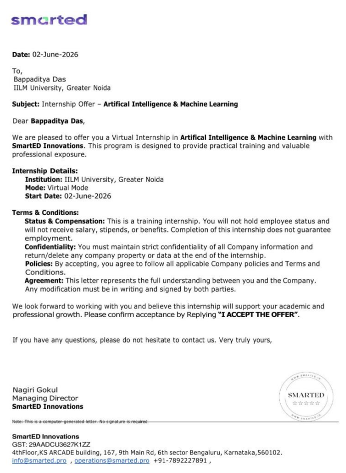
| Module | Module Title | Key Topics Covered | Key Learning Outcome |
|---|---|---|---|
| | | • Problem Formulation & State Space: Foundations<br>• Uninformed Search Algorithms: Foundations and Analysis<br>• Informed Search Algorithms: Navigating AI's Decision Pathways<br>• Search Limitations & Comparisons | |
| Module 3 | Rule-Based Systems and Traditional Machine Learning | • Introduction to Rule-Based Systems<br>• Advanced Concepts and Components of Expert Systems<br>• Supervised Learning Algorithms: Regression and Classification Essentials<br>• Model Evaluation Metrics: Confusion Matrix, Precision, Recall, F1-Score, ROC-AUC<br>• Overfitting and Regularization: Mastering the Bias-Variance Tradeoff<br>• Building an End-to-End Machine Learning Pipeline: From Raw Data to Reliable Models | Gained hands-on exposure to rule-based reasoning, core supervised ML algorithms, and rigorous model evaluation practices. |
| Module 4 | Unsupervised Learning and Probabilistic Reasoning | • Clustering Techniques: K-Means and Hierarchical Clustering Explained<br>• Dimensionality Reduction: The Hidden Structures Unveiled<br>• Probabilistic Reasoning: Foundations and Bayes' Theorem<br>• Naive Bayes Classifier: Assumptions and Text Classification Use Case | Understood unsupervised learning techniques and probabilistic approaches for reasoning and classification under uncertainty. |

| Module | Module Title | Key Topics Covered | Key Learning Outcome |
|---|---|---|---|
| | | • Bayesian Networks: Modeling Uncertainty with Graphical Models | |
| Module 5 | Deep Learning and Computer Vision | • Neural Network Foundations: From Biology to Modern AI<br>• Backpropagation and Training: How Neural Networks Learn<br>• Introduction to TensorFlow, Keras, and PyTorch | Developed a working understanding of neural network fundamentals and practical exposure to leading deep learning frameworks. |
| Module 6 | NLP, Generative AI and Industry Applications | • Text Preprocessing: The Essential First Step in NLP<br>• Vectorization Techniques in Natural Language Processing<br>• Sentiment Analysis Pipeline Using Scikit-learn<br>• Generative AI Introduction: Transformers, GPT, and Prompt Engineering | Learned core NLP preprocessing and vectorisation techniques, and explored Generative AI concepts and industry use cases. |

### 4.11 Certificates of Completion (Annexure at a Glance)

The Internship Offer Letter and the Certificate of Internship Completion issued by SmartED Innovations are attached in full in the preceding section (Internship Completion Certificate) of this report as Annexure I and Annexure II respectively. A compact, at-a-glance view of both documents is provided below for quick reference.

**Table 4.2: Internship Offer Letter and Completion Certificate – Annexure Summary**



smarted

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**Internship Details:**
**Institution:** IILM University, Greater Noida
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**Terms & Conditions:**
**Status & Compensation:** This is a training internship. You will not hold employee status and will not receive salary, stipends, or benefits. Completion of this internship does not guarantee employment.
**Confidentiality:** You must maintain strict confidentiality of all Company information and return/delete any company property or data at the end of the internship.
**Policies:** By accepting, you agree to follow all applicable Company policies and Terms and Conditions.
**Agreement:** This letter represents the full understanding between you and the Company. Any modification must be in writing and signed by both parties.

We look forward to working with you and believe this internship will support your academic and professional growth. Please confirm acceptance by Replying **"I ACCEPT THE OFFER"**.

If you have any questions, please do not hesitate to contact us. Very truly yours,

Nagiri Gokul
Managing Director
**SmartED Innovations**

SMARTED

~~Note: This is a computer-generated letter. No signature is required~~

**SmartED Innovations**
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info@smarted.pro , operations@smarted.pro +91-7892227891 ,

**Annexure I: Internship Offer Letter**

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**CERTIFICATE**
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Bappaditya Das

Has successfully completed training and an internship with **SmartED Innovations** in,
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Throughout this internship, the student has consistently shown exceptional enthusiasm, dedication, and a strong willingness to learn and contribute

**STUDENT ID:** AIIC1327

**Date:** 11 Jul 2026

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**SAI SREEKAR**
Director & co-founder

**USNG LEARNSPACE EDTECH PVT LTD** www.smarted.pro

**Annexure II: Certificate of Completion**

## 5. Bibliography/References


[1] Pedregosa, F. et al., “Scikit-learn: Machine Learning in Python,” Journal of Machine Learning Research, vol. 12, pp. 2825–2830, 2011.

[2] McKinney, W., “Data Structures for Statistical Computing in Python,” Proceedings of the 9th Python in Science Conference, 2010. (Pandas library documentation, pandas.pydata.org)

[3] Scikit-learn Developers, “Scikit-learn User Guide,” https://scikit-learn.org/stable/user_guide.html

[4] NYC Airbnb Open Data, publicly available listings dataset used for the room classification task.

[5] Hunter, J.D., “Matplotlib: A 2D Graphics Environment,” Computing in Science & Engineering, vol. 9, no. 3, pp. 90–95, 2007.

[6] SmartED Innovations, Internship Program Materials and Session Notes, Artificial Intelligence Virtual Internship, 2025–26.